

LPDDR4X Datasheet

RS256M32LS4D4BNR-53BT
200-ball

Revision 1.0

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Revision History

Version	Date	Changes
V1.0	2025-8-10	Basic spec and architecture

Notes This data sheet contains minimum and maximum limits specified over the power supply and temperature range set forth herein. Although considered final, these specifications are subject to change at any time without notice, as further product development and data characterization sometimes occur.

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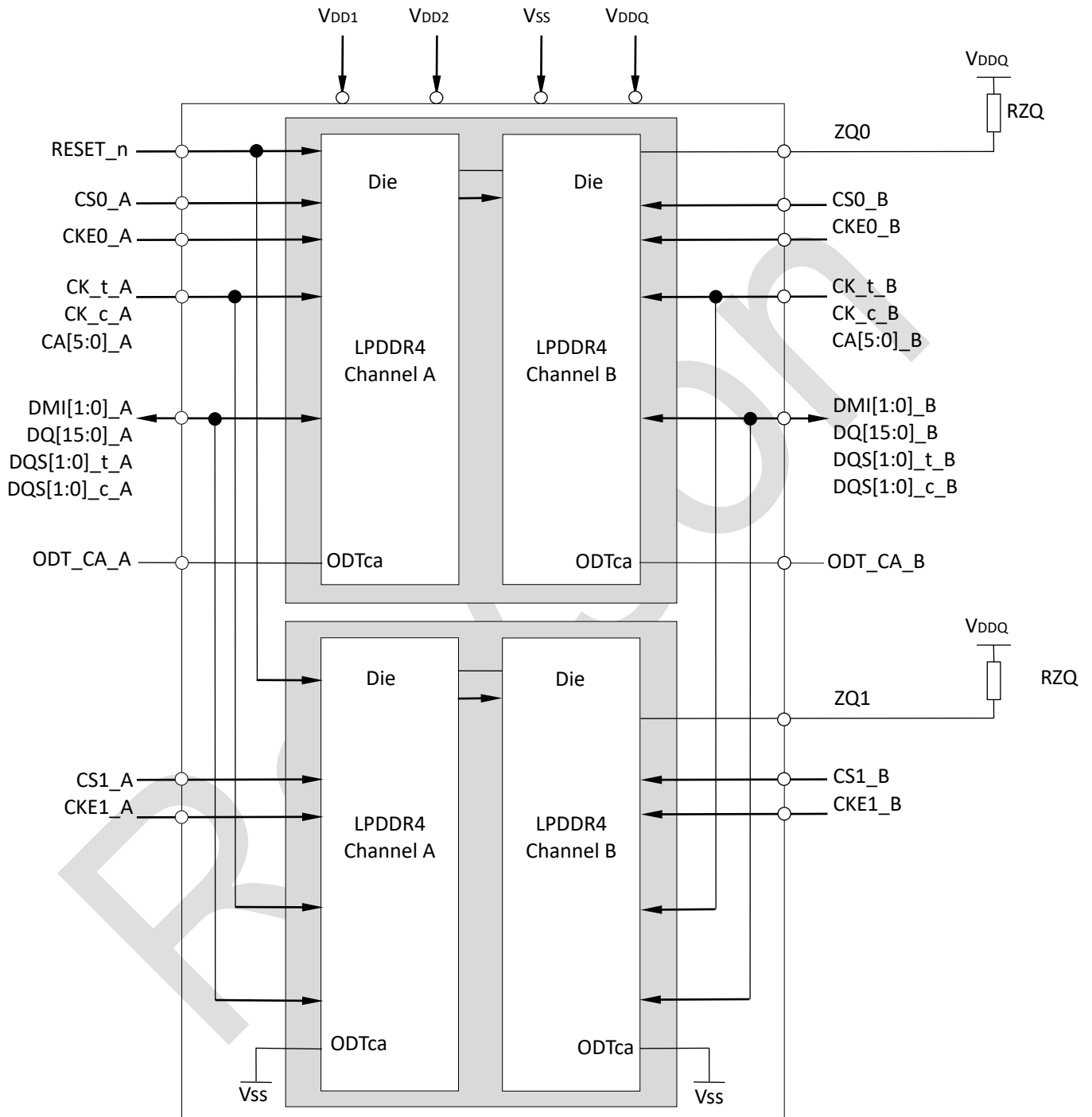
1. Product Overview

1.1. Feature Overview

- Ultra-low-voltage core and I/O power supplies
 - V_{DD1} = 1.70-1.95V; 1.80V nominal
 - V_{DD2} = 1.06-1.17V; 1.10V nominal
 - V_{DDQ} = 0.57-0.65V; 0.60V nominal
- Frequency range
 - 1866-10 MHz (data rate range per pin:3733-20Mbps/s)
- 16n prefetch DDR architecture
- 8 internal banks per channel for concurrent operation
- Single-data-rate CMD / ADR entry
- Bidirectional / differential data strobe per byte lane
- Programmable READ and WRITE latencies (RL / WL)
- Programmable and on-the-fly burst lengths (BL = 16, 32)
- Directed per-bank refresh for concurrent bank operation and ease of command scheduling
- Up to 8.53 GB / s per die x16 channel
- On-chip temperature sensor to control self refresh rate
- Partial-array self refresh (PASR)
- Selectable output drive strength (DS)
- Clock-stop capability
- RoHS-compliant, “green” packaging
- $V_{DD1}/V_{DD2}/V_{DDQ}$: 1.80V/1.10V/0.60V
- Array configuration
 - 256Meg x 32 (2 channels x 16 I/O)
 - 128M16 x 4 die in package
- FBGA “green” package
 - 200-ball FBGA (10mm x15mm x1.0mm Max)
- Speed grade,cycle time
 - 535ps@ RL = 32/36
- Operating temperature range
 - -25°C to + 85°C

2. Physical Specifications

2.1. Function Block Diagram



Quad-Die, Dual-Channel, Dual-Rank Package Block Diagram

2.2. Package ballout & Addressing

200-Ball Dual-Channel, Dual -Rank Discrete FBGA

	1	2	3	4	5	6	7	8	9	10	11	12
A	DNU	DNU	V _{SS}	V _{DD2}	ZQ0			ZQ1	V _{DD2}	V _{SS}	DNU	DNU
B	DNU	DQ0_A	V _{DDQ}	DQ7_A	V _{DDQ}			V _{DDQ}	DQ15_A	V _{DDQ}	DQ8_A	DNU
C	V _{SS}	DQ1_A	DMI0_A	DQ6_A	V _{SS}			V _{SS}	DQ14_A	DMI1_A	DQ9_A	V _{SS}
D	V _{DDQ}	V _{SS}	DQS0_t_A	V _{SS}	V _{DDQ}			V _{DDQ}	V _{SS}	DQS1_t_A	V _{SS}	V _{DDQ}
E	V _{SS}	DQ2_A	DQS0_c_A	DQ5_A	V _{SS}			V _{SS}	DQ13_A	DQS1_c_A	DQ10_A	V _{SS}
F	V _{DD1}	DQ3_A	V _{DDQ}	DQ4_A	V _{DD2}			V _{DD2}	DQ12_A	V _{DDQ}	DQ11_A	V _{DD1}
G	V _{SS}	ODT_CA_A	V _{SS}	V _{DD1}	V _{SS}			V _{SS}	V _{DD1}	V _{SS}	NC	V _{SS}
H	V _{DD2}	CA0_A	CS1_A	CS0_A	V _{DD2}			V _{DD2}	CA2_A	CA3_A	CA4_A	V _{DD2}
J	V _{SS}	CA1_A	V _{SS}	CKE0_A	CKE1_A			CK_t_A	CK_c_A	V _{SS}	CA5_A	V _{SS}
K	V _{DD2}	V _{SS}	V _{DD2}	V _{SS}	NC			NC	V _{SS}	V _{DD2}	V _{SS}	V _{DD2}
L												
M												
N	V _{DD2}	V _{SS}	V _{DD2}	V _{SS}	NC			NC	V _{SS}	V _{DD2}	V _{SS}	V _{DD2}
P	V _{SS}	CA1_B	V _{SS}	CKE0_B	CKE1_B			CK_t_B	CK_c_B	V _{SS}	CA5_B	V _{SS}
R	V _{DD2}	CA0_B	CS1_B	CS0_B	V _{DD2}			V _{DD2}	CA2_B	CA3_B	CA4_B	V _{DD2}
T	V _{SS}	ODT_CA_B	V _{SS}	V _{DD1}	V _{SS}			V _{SS}	V _{DD1}	V _{SS}	RESET_n	V _{SS}
U	V _{DD1}	DQ3_B	V _{DDQ}	DQ4_B	V _{DD2}			V _{DD2}	DQ12_B	V _{DDQ}	DQ11_B	V _{DD1}
V	V _{SS}	DQ2_B	DQS0_c_B	DQ5_B	V _{SS}			V _{SS}	DQ13_B	DQS1_c_B	DQ10_B	V _{SS}
W	V _{DDQ}	V _{SS}	DQS0_t_B	V _{SS}	V _{DDQ}			V _{DDQ}	V _{SS}	DQS1_t_B	V _{SS}	V _{DDQ}
Y	V _{SS}	DQ1_B	DMI0_B	DQ6_B	V _{SS}			V _{SS}	DQ14_B	DMI1_B	DQ9_B	V _{SS}
AA	DNU	DQ0_B	V _{DDQ}	DQ7_B	V _{DDQ}			V _{DDQ}	DQ15_B	V _{DDQ}	DQ8_B	DNU
AB	DNU	DNU	V _{SS}	V _{DD2}	V _{SS}			V _{SS}	V _{DD2}	V _{SS}	DNU	DNU
	1	2	3	4	5	6	7	8	9	10	11	12

Top View

LPDDR4_A(Channel A)
 LPDDR4_B(Channel B)
 ZQ,ODT_CA,RESET
 Supply
 Ground

2.3. Pad Definition

“_A” and “_B” indicate DRAM channels. “_A” pads are present in all devices while “_B” pads are present in dual channel SDRAM devices only.

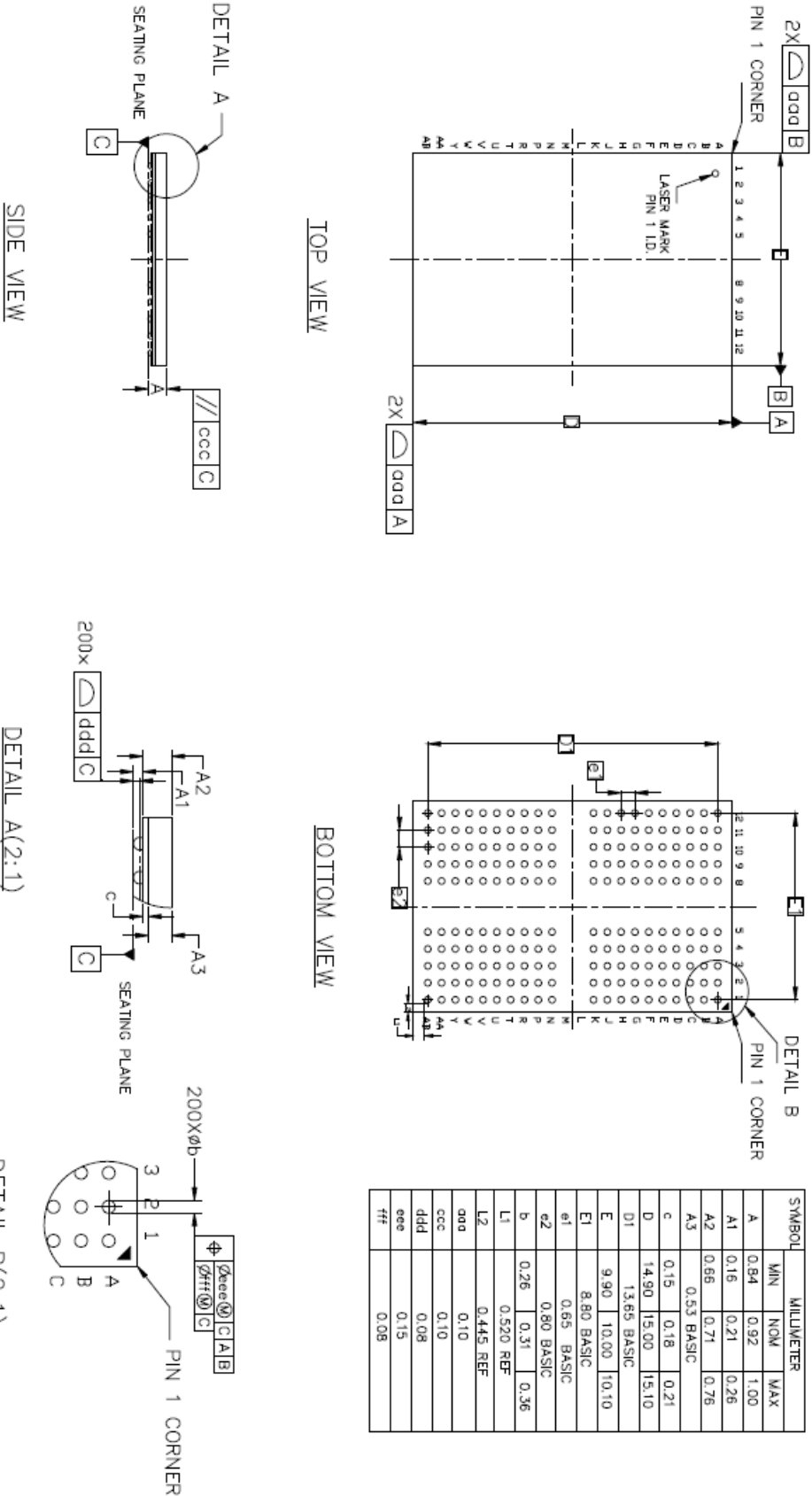
LPDDR4X pad definitions are the same as LPDDR4, except ODT_CA pins as described in the following Table

Symbol	Type	Description
CK_t_A, CK_c_A CK_t_B, CK_c_B	Input	Clock: CK_t and CK_c are differential clock inputs. All address, command and control input signals are sampled on positive edge of CK_t and the negative edge of CK_c.AC timings for CA parameters are referenced to clock. Each channel (A, B) has its own clock pair.
CKE0_A, CKE1_A CKE0_B, CKE1_B	Input	Clock enable: CKE HIGH activates and CKE LOW deactivates the internal clock signals, input buffers, and output drivers. Power-saving modes are entered and exited via CKE transitions. CKE is sampled at the rising edge of CK.
CS0_A, CS1_A CS0_B, CS1_B	Input	Chip select: Each channel (A, B) has its own CS signals.
CA[5:0]_A CA[5:0]_B	Input	Command/address inputs: Provide the command and address inputs according to the command truth table. Each channel (A, B) has its own CA signals.
ODT_CA_A ODT_CA_B	Input	CA ODT Control: The ODT_CA pin is ignored by LPDDR4X devices. CA ODT is fully controlled through MR11 and MR22. The ODT_CA pin shall be connected to a valid logic level.
DQ[15:0]_A DQ[15:0]_B	I/O	Data input/output: Bidirectional data bus.
DQS[1:0]_t_A DQS[1:0]_c_A DQS[1:0]_t_B DQS[1:0]_c_B	I/O	Data strobe: DQS_t and DQS_c are bidirectional differential output clock signals used to strobe data during a READ or WRITE. The data strobe is generated by the DRAM for a READ and is edge-aligned with data. The data strobe is generated by the SoC memory controller for a WRITE and is trained to precede data. Each byte of data has a data strobe signal pair. Each channel (A, B) has its own DQS_t and DQS_c strobes.
DMI[1:0]_A DMI[1:0]_B	I/O	Data mask/Data bus inversion: DMI is a dual use bidirectional signal used to indicate data to be masked, and data which is inverted on the bus. For data bus inversion(DBI), the DMI signal is driven HIGH when the data on the data bus is inverted, or driven LOW when the data is in its normal state. DBI can be disabled via a mode register setting. For data mask, the DMI signal is used in combination with the data lines to indicate data to be masked in a MASK WRITE command (see the Data Mask (DM) and Data Bus Inversion (DBI) sections for details). The data mask function can be disabled via a mode register setting. Each byte of data has a DMI signal. Each channel has its own DMI signals.

Symbol	Type	Description
ZQ0, ZQ1	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin shall be connected to V _{DDQ} through a 240Ω ±1% resistor.
V _{DDQ} , V _{DD1} , V _{DD2}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V _{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	RESET: When asserted LOW, the RESET pin resets all channels of the die.
DNU	–	Do not use: Must be grounded or left floating.
NC	–	No connect: Not internally connected.

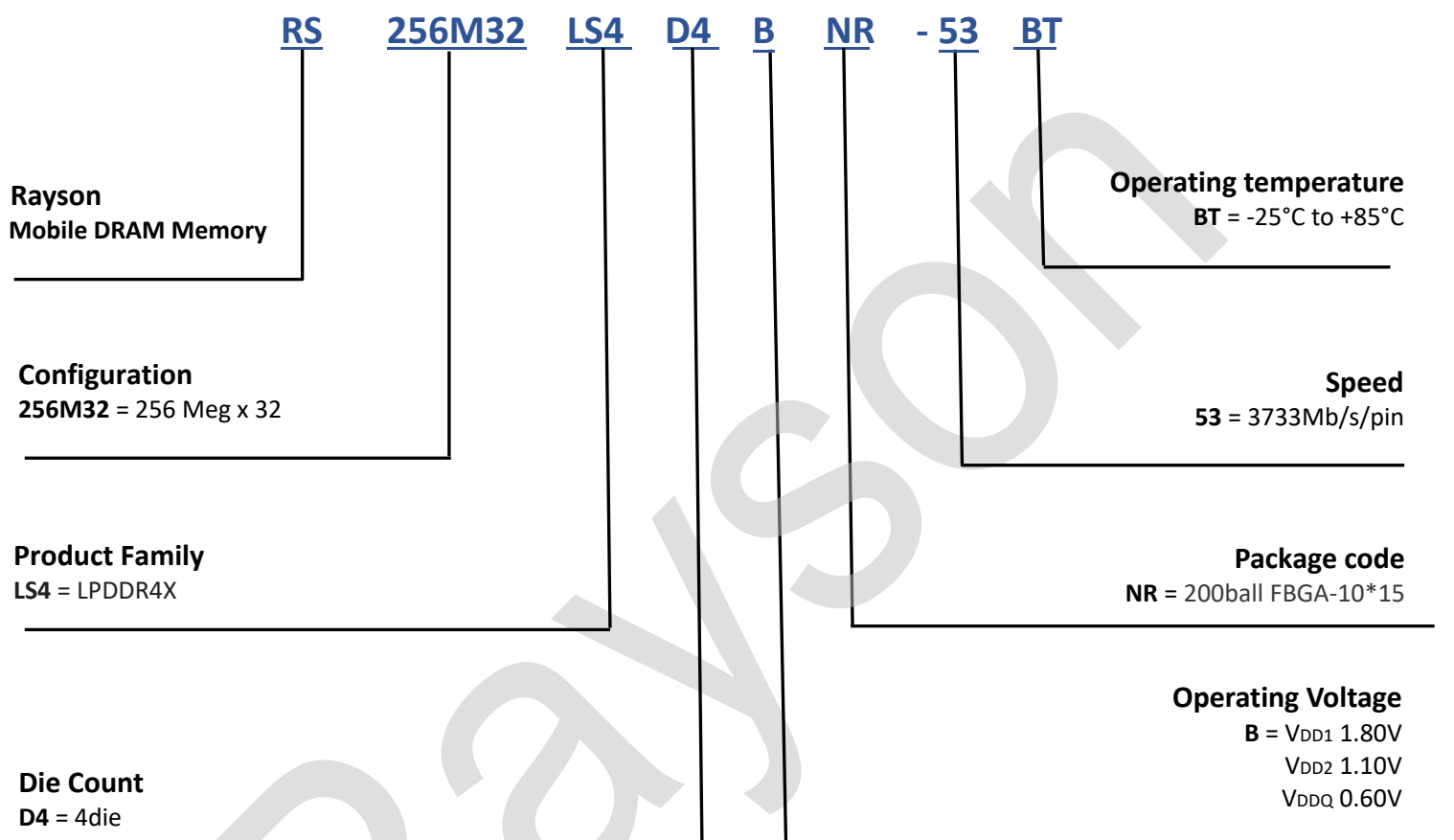
2.4. Discrete Package Dimension

10mm X15mm



3. Core Specifications

3.1. Part Number Decoding



3.2. Ordering Options

Table 1: Key Timing Parameters

Speed Grade	Clock Rate (MHz)	Data Rate (Mb/s/pin)	WRITE Latency		READ Latency	
			Set A	Set B	DBI Disabled	DBI Enabled
-53	1866	3733	16	30	32	36

Table 2: Part Number List

Part Number	Total Density	Data Rate	Operating temperature
RS256M32LS4D4BNR-53BT	1GB(8Gb)	3733 Mb/s/pin	-25°C to + 85°C

Table 3: Refresh Requirement Parameters

Parameter	Symbol	2Gb Single Channel Die	unit
REFRESH cycle time (all banks)	t_{RFCab}	130	ns
REFRESH cycle time (per bank)	t_{RFCpb}	60	ns
Per bank refresh to per bank refresh time (different bank)	$t_{PBR2PBR}$	60	ns

3.3. Die Addressing Table

Configuration		256M32 (8Gb/package)
Die Configuration	Channel A, rank 0	x16 mode × 1 die
	Channel B, rank 0	x16 mode × 1 die
	Channel a, rank 1	x16 mode × 1 die
	Channel B, rank 1	x16 mode × 1 die
Die Addressing	Memory density (per die)	2Gb
	Memory density (per channel)	4Gb
	Configuration	16Mb × 16 DQ × 8 banks x 2channels x 2 ranks
	Number of channels (per die)	1
	Number of ranks per channel	1
	Number of banks (per channel)	8
	Array prefetch (bits, per channel)	256
	Number of rows (per channel)	16,384
	Number of columns (fetch boundaries)	64
	Page size (bytes)	2048
	Channel density (bits per channel)	2,147,483,648
	Total density (bits per die)	2,147,483,648
	Bank address	BA[2:0]
	Row address	R[13:0]
	Column address	C[9:0]
	Burst starting address boundary	64-bit

Notes: 1、Refer to Package Block Diagrams section in Product specification and SDRAM Addressing Section in General LPDDR4X specification.

3.4. Mode Register Contents

MR0		
OP7		OP[0] = 0b: Both legacy and modified refresh mode supported
OP6		
OP5		
OP4		
OP3		
OP2		
OP1		
OP0	REF	

MR3		
OP7		OP[2] = 0b: PPR protection disabled (default) 1b: Reserved
OP6		
OP5		
OP4		
OP3		
OP2	PPRP	
OP1		
OP0		

MR5		
OP7	Manufacturer ID	0000 0001b : Samsung
OP6		
OP5		
OP4		
OP3		
OP2		
OP1		
OP0		

MR6		
OP7	Revision ID1	0000 0101b
OP6		
OP5		
OP4		
OP3		
OP2		
OP1		
OP0		

MR8		
OP7	I/O width	OP[7:6] = 00b: x16/channel
OP6		
OP5	Density	OP[5:2] = 0000b: 2Gb single channel die
OP4		
OP3		
OP2		
OP1		
OP0		

MR13		
OP7		OP[2]=0b: Normal operation (default) 1b: Output the V _{REF(CA)} value on DQ7 and V _{REF(DQ)} value on DQ6
OP6		
OP5		
OP4		
OP3		
OP2	VR0	
OP1		
OP0		

Notes: 1、 *The contents of MR0, MR[6:3], MR8, MR13 will reflect information specific to each in these package*

2、 *Other bits not defined above and other mode registers are referred to in Mode Register Assignments and Definitions section.*

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4. IDD Parameters

Symbol		Power Supply	2Gb (x16/Ch, 1-Chip)	Units
			3733Mbps	
IDD0	IDD0 ₁	VDD1	5	mA
	IDD0 ₂	VDD2	25.25	mA
	IDD0 _Q	VDDQ	0.1	mA
IDD2P	IDD2P ₁	VDD1	0.5	mA
	IDD2P ₂	VDD2	1.0	mA
	IDD2P _Q	VDDQ	0.1	mA
IDD2PS	IDD2PS ₁	VDD1	0.5	mA
	IDD2PS ₂	VDD2	1.0	mA
	IDD2PS _Q	VDDQ	0.1	mA
IDD2N	IDD2N ₁	VDD1	0.5	mA
	IDD2N ₂	VDD2	9.25	mA
	IDD2N _Q	VDDQ	0.1	mA
IDD2NS	IDD2NS ₁	VDD1	0.5	mA
	IDD2NS ₂	VDD2	6.75	mA
	IDD2NS _Q	VDDQ	0.1	mA
IDD3P	IDD3P ₁	VDD1	0.5	mA
	IDD3P ₂	VDD2	3.25	mA
	IDD3P _Q	VDDQ	0.1	mA
IDD3PS	IDD3PS ₁	VDD1	0.5	mA
	IDD3PS ₂	VDD2	3.25	mA
	IDD3PS _Q	VDDQ	0.1	mA
IDD3N	IDD3N ₁	VDD1	0.5	mA
	IDD3N ₂	VDD2	10.25	mA
	IDD3N _Q	VDDQ	0.1	mA
IDD3NS	IDD3NS ₁	VDD1	0.5	mA
	IDD3NS ₂	VDD2	7.75	mA
	IDD3NS _Q	VDDQ	0.1	mA
IDD4R	IDD4R ₁	VDD1	0.55	mA
	IDD4R ₂	VDD2	185	mA
	IDD4R _Q	VDDQ	97.5	mA
IDD4W	IDD4W ₁	VDD1	0.5	mA
	IDD4W ₂	VDD2	175	mA
	IDD4W _Q	VDDQ	0.1	mA
IDD5	IDD5 ₁	VDD1	22.5	mA
	IDD5 ₂	VDD2	80.25	mA
	IDD5 _Q	VDDQ	0.1	mA
IDD5AB	IDD5AB ₁	VDD1	1	mA
	IDD5AB ₂	VDD2	11.25	mA
	IDD5AB _Q	VDDQ	0.1	mA

Symbol			Power Supply	2Gb (x16/Ch, 1-Chip)	Units
				3733Mbps	
IDD5PB	IDD5PB ₁		VDD1	1	mA
	IDD5PB ₂		VDD2	11.25	mA
	IDD5PB _Q		VDDQ	0.1	mA
IDD6	IDD6 ₁	25°C	VDD1	0.11	mA
		85°C		0.4	
	IDD6 ₂	25°C	VDD2	0.12	mA
		85°C		1.2	
	IDD6 _Q	25°C	VDDQ	0.025	mA
		85°C		0.035	